Lecture 3

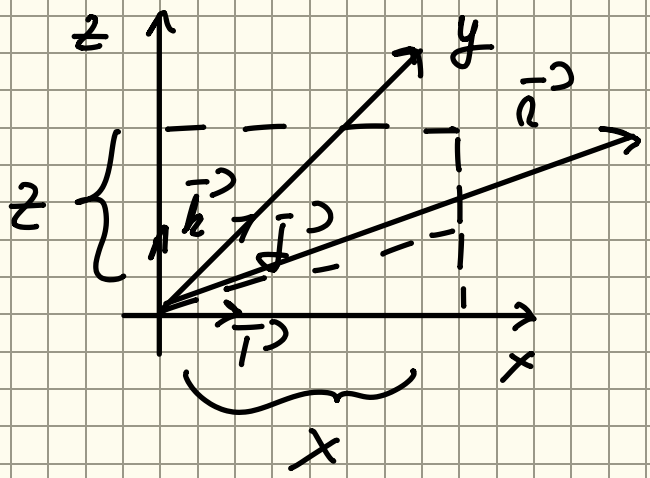
I Kinematics

1. The elements of motion

1.1. The reference frame

1.2. The trajectory

1.3. The position vector

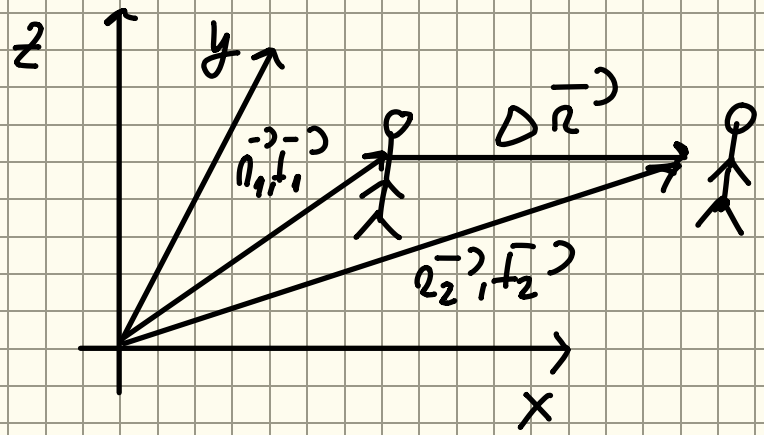


$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$r = |\vec{r}| = \sqrt{\vec{r}^2}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

1.4. The displacement vector



$$\vec{r}_2 = \vec{r}_1 + \Delta\vec{r}$$

$$\Rightarrow \Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

↑ displacement vector

$\Delta\vec{r} \neq 0 \Rightarrow$ motion

2. Velocity

$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$
$$\Delta t = t_2 - t_1$$

$$\Rightarrow \vec{v}_r = \frac{\Delta\vec{r}}{\Delta t} \rightarrow \text{the average velocity}$$

↑ velocity

speed = (velocity)

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \qquad \vec{v} \overset{dif}{:=} \frac{d\vec{r}}{dt}$$

3. The rectilinear motion with constant velocity:

$\vec{v} \to$ v - const

$\vec{r} \to$ s (space, length of time)

! $s = f(t)$

$$v = \frac{ds}{dt} \Rightarrow ds = v \cdot dt \;/\; \int$$

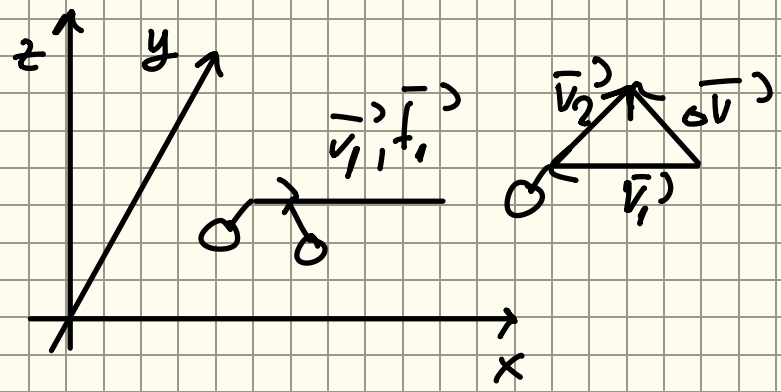
$$\int_{s(0)}^{s(t)} ds = \int_0^t v \cdot dt'$$

$$\Leftrightarrow s(t) - s(0) = v \cdot t' \Big|_0^t \Rightarrow$$

$$\boxed{s(t) = s(0) + vt}$$

Law of motion

4. The acceleration



$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

$$\Delta t = t_2 - t_1$$

$$\vec{a}_n = \frac{\Delta\vec{v}}{\Delta t} \text{ - average acc.}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \qquad \vec{a} \overset{dif}{:=} \frac{d\vec{v}}{dt} \Rightarrow \text{the acceleration}$$

5. The rectilinear motion with constant acceleration.

$\vec{a} \to a$ - constant
$\vec{v} \to v$
$\vec{r} \to r$
$r = f(t)$!

$$a = \frac{dv}{dt} \Rightarrow dv = a \cdot dt \;/\; \int$$

$$\int_{v(0)}^{v(t)} dv = \int_0^t a \cdot dt' \Rightarrow$$

$$v\Big|_{v(0)}^{v(t)} = a \cdot t'\Big|_0^t \Rightarrow v(t) - v(0) = a(t-0)$$

$$\Rightarrow \boxed{v(t) = v(0) + at}$$

Law of velocity

$$v = \frac{dr}{dt} \Rightarrow dr = v \cdot dt \;/\; \int$$

$$\int_{r(0)}^{r(t)} dr = \int_0^t v(t)\,dt' = \int_0^t [v(0) - at']\,dt'$$

$$r\Big|_{r(0)}^{r(t)} = \int_0^t v(0)\,dt' + \int_0^t at'\,dt'$$

$$r(t) - r(0) = v(0)\int_0^t dt' + a\int_0^t t'\,dt'$$

$$r(t) - r(0) = v(0) \cdot t'\Big|_0^t - a \cdot \frac{t'^2}{2}\Big|_0^t = v(0)t + a\frac{t^2}{2}$$

$$\boxed{r(t) = r(0) + v(0)t + a\frac{t^2}{2}}$$

Laws of motion

6. The Laws of Galilei

$s(t) - s(0) = s$

$v(0) \to v_0$

$v(t) \to v$

Law of velocity $\Rightarrow t = \frac{v - v_0}{a}$

$$s = v_0 \frac{v - v_0}{a} + \frac{a}{2} \frac{(v - v_0)^2}{a^2}$$

$$s = \frac{2v_0 v - 2v_0^2 + v^2 + v_0^2 - 2vv_0}{2a}$$

$$s = \frac{v^2 - v_0^2}{2a} \Rightarrow v^2 - v_0^2 = 2sa$$

$$\boxed{v^2 = v_0^2 + 2sa} \to \text{Law of Galilei}$$

7. Examples

$$s(t) = 3t^5 + \frac{1}{\sqrt{t}} e^{-2t} + \sin(4\pi t)$$

$$v(t) = \frac{d}{dt}\left[3t^5 + t^{-\frac{1}{2}} + e^{-2t} + \sin(4\pi t)\right]$$

$$v(t) = 15t^4 - \frac{1}{2\sqrt{t^3}} - 2e^{-2t} + 4\pi \cos(4\pi t)$$

$$a(t) = \frac{d}{dt}\left[15t^4 - \frac{1}{2}t^{-\frac{3}{2}} - 2e^{-2t} + 4\pi \cos(4\pi t)\right]$$

$$a(t) = 60t^3 + \frac{3}{4}t^{-\frac{5}{2}} + 4e^{-2t} - 16\pi^2 \sin(4\pi t)$$

$$p \xrightarrow{\frac{d}{dt}} v \xrightarrow{\frac{d}{dt}} a$$

$$a \xrightarrow{S} v \xrightarrow{S} p$$